

Shakra Batool

CONTACT INFORMATION	shakra.batool123@gmail.com (+92)335-9650799
RESEARCH INTERESTS	• Machine Learning and Deep Learning for biomedical and clinical applications. • Data Analysis and Statistical Modeling of healthcare and biological datasets for trend analysis, and predictive insights. • Large Language Models (LLMs) for the automation and validation of data extraction, analytical tasks, and decision-making in biomedical datasets • Computational Informatics for integrating and analyzing large-scale biomedical data to support precision medicine • Medical Image Processing (MRI, CT, X-rays, Histopathology) for disease detection and segmentation. • Generative AI for data generation and augmentation.
EDUCATION	National University of Science and Technology (NUST), Islamabad, Pakistan. Masters in Bioinformatics Sep. 2021 - August 2023 CGPA: 3.85/4.0 • Relevant Courses: Applied Machine Learning, Data Analysis and Statistics, Deep Learning, Deep Learning in Medicine COMSATS University., Islamabad, Pakistan. Bachelors in Bioinformatics Feb. 2017 - Jan 2021 CGPA: 3.51/4.0 • Relevant Courses: Artificial Intelligence and Neural Networks, Bioinformatics Analysis
WORK EXPERIENCE	• Software Developer Jan. 2025 – Present <i>Codeaza Technologies, Rawalpindi, Pakistan</i> • Led the design and development of full-stack web applications and AI-enhanced platforms using React, Flask, and Python. • Built scalable microservice architectures and integrated NLP libraries (e.g., spaCy) for sentiment analysis and context-aware recommendations across domains such as hospitality and healthcare. • Collaborated with UI/UX designers, backend engineers, and stakeholders to deliver customized solutions that support data-driven decision-making. • Contributed to AI feature development for dashboards, enhancing system performance, reliability, and user engagement. • Research Associate Sep. 2023 – Dec. 2024 <i>Image Analysis Lab (SINES), National University of Sciences and Technology (NUST)</i> • Focused on cutting-edge medical image analysis, automated segmentation, and cancer detection in 3D MRI and CT data. • Designed deep learning pipelines utilizing 3D U-Net with multi-head attention mechanisms, incorporated preprocessing techniques such as cropping, intensity normalization, and spatial resampling to improve segmentation accuracy and clinical applicability.

- Built a web-based Hospital Management System (HMS) embedded with LLM capabilities using Gemini for automatic summarization and analysis of patient medical records.
 - Performed experimental evaluations and result visualization to validate models against benchmark datasets.
 - Organized clinical annotation meetups with radiologists to facilitate expert labeling of medical images and ensure clinically aligned model training.
 - Organized and conducted workshops on “Machine Learning in Biology” to educate doctors and researchers about AI-driven medical imaging and data analysis techniques.
 - Used Roboflow for object detection, dataset annotation, augmentation, and pose estimation tasks in both clinical and non-clinical domains.
 - Developed YOLO based object detection models to identify multiple entities in dynamic scenes, such as golf ball tracking and multi-class detection including humans, grass, and sky.
 - Mentored final-year undergraduate students on AI/ML research projects, supporting experimental design, model training, and paper writing.
- **Research Assistant** Sep. 2020 – Jan. 2021
Medical Image and Diagnostic Lab, National Centre for Artificial Intelligence (NCAI)
- Contributed to AI-based radiological image classification, focusing on disease diagnosis using convolutional neural networks (CNNs).
 - Curated and annotated diagnostic datasets (X-ray, CT) using libraries like OpenCV to construct high-quality, task-specific training data.
 - Designed and implemented CNN architectures for multi-class classification tasks, optimizing for diagnostic accuracy and generalization.
 - Collaborated closely with radiologists to interpret model outputs, ensuring alignment of predictions with clinical ground truth, bridging research outcomes with real-world applications.

RESEARCH PROJECTS

- **Development of Deep Learning Pipeline for Airways Segmentation in Human Lungs** *Approach:* This project focused on the segmentation of airways using 3D CT scans of human lungs. The dataset was obtained from an online MICCAI competition. A comprehensive preprocessing pipeline was implemented, which included windowing, filtering, thresholding, and resizing of the CT images. The 3D volumes were then converted into 2D slices to make them compatible with standard convolutional architectures. The prepared dataset was fed into a U-Net deep neural network for training. High-Performance Computing (HPC) clusters were used to accelerate the training process. To improve segmentation accuracy, hyperparameter tuning was carried out.
Outcome: A trained U-Net model capable of effectively segmenting airway structures from 2D DICOM slices of lung CT scans.
- **Cancer Detection Using Convolutional Neural Networks** *Approach:* The objective of this project was to detect tumors from both mammograms and histopathology images. Several CNN models—VGG16, ResNet50, and MobileNet were evaluated and compared based on their performance. The CBIS-DDSM dataset, a modern version of the Digital Database for Screening Mammography (DDSM), was used for model training and validation. Preprocessing steps used in the project included normalization, resizing, and augmentation to enhance generalization. Each model was trained and validated to classify tumors, and their comparative performance was assessed based on accuracy and inference time.
Outcome: The MobileNet architecture resulted as the best-performing model, providing accurate classification of tumor types with minimal computational cost.
- **Cancer Segmentation Using U-Net** *Approach:* This project involved the segmentation of cancer cell nuclei from microscopic images using the U-Net

architecture. The dataset used was the publicly available Data Science Bowl 2018 dataset, which included images with masks representing nuclei. To enhance model robustness and reduce overfitting, data augmentation techniques were applied, including rotation and flipping. The U-Net model was trained to learn pixel-wise segmentation, capturing both the shape and boundaries of the nuclei accurately.

Outcome: The resulting trained U-Net model demonstrated strong performance in segmenting nuclei from biomedical images and was capable of handling complex and overlapping structures.

OTHER PROJECTS

- **Breast Tumor Classification App** Developed an application for classifying breast tumors using radiographic images. Utilized convolutional neural networks (CNNs) to assist in early and automated diagnosis.
- **Diabetes Prediction System** Built a machine learning-based solution for early-stage diabetes detection using patient medical features. Achieved high classification accuracy with models such as Random Forest and Support Vector Machine (SVM).
- **AI-Enhanced Hospital Management System** Designed a hospital management platform integrating AI capabilities for automated lab report analysis and abnormal result detection.
- **Pneumonia Detection Using Chest X-rays** Implemented a deep learning pipeline using CNNs to detect pneumonia from chest radiographs. Improved classification robustness through data augmentation and hyperparameter tuning.
- **Cow Disease Prediction Model** Created predictive models for cow disease detection using various machine learning algorithms, including Multi-Layer Perceptron (MLP). Conducted performance analysis using metrics such as accuracy, F1-score, and AUC.
- **Golf Ball Tracking using YOLO v8** Designed and implemented a real-time object detection system using YOLO v8 to track golf ball trajectory and identify multiple objects in the scene, including players (humans), grass, sky, and surrounding elements. Collected and annotated a custom video dataset, used the model YOLO v8 for multi-class object detection, and evaluated its performance based on accuracy, speed, and robustness under various environmental conditions.
- **Roboflow for Object Detection, Data Annotation, and Pose Estimation** Utilized Roboflow for efficient data annotation, augmentation, object detection, and pose estimation tasks. Created and labeled custom datasets of traffic signals and cars, applied preprocessing pipelines, and exported data in YOLO-compatible formats to train deep learning models. Also used pose estimation capabilities within Roboflow to track and analyze keypoints for sportsmen and object posture recognition.

LANGUAGES AND TOOLS

- **Languages:** Python, Java, C++, MySQL, R, Matlab, HTML, C#, and React.
- **Tools:** Eclipse, Idle, Google Colab, Netbeans, VS Code, Git, SPSS, Linux, and Docker.

OTHER SKILLS

- Shell scripting
- Deep Learning Libraries:
 - TensorFlow, PyTorch.
- App development:
 - MySQL, React, Express.

PUBLICATIONS

- Glioblastomas segmentation using Deep Learning BraTS 2023 Challenge
(Submitted in Synapse competition)

TEACHING EXPERIENCE

- Conducted Python labs for undergraduate students.
- Conducted workshop on "Machine Learning-The Future of Biology" as instructor.
- Organized internships focusing on artificial intelligence and machine learning.

VOLUNTEER WORK

- Fund-raising and distribution to local deserving people affected by COVID-19.
- Awareness talk on Muscular Dystrophy and Thalassymia.
- Hospital visit to understand the causes and effects of Muscular Dystrophy and Thalassymia in young children.
- Hospital visit to understand the role of airway anatomy in Chronic Obstructive Pulmonary Diseases(COPD).